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A 35-year-old female patient presents to the gynecology clinic with complaints of abdominal pain and a palpable mass in her lower abdomen. On physical examination, a centrally located, smooth-surfaced abdominopelvic mass measuring approximately 16 weeks' gestational size is palpated. What is the most likely diagnosis based on the clinical presentation?

- ☐ Ovarian tumor
- ☐ Molar pregnancy
- ☒ Myoma
- ☐ Ectopic pregnancy

Explanation :

Myoma (uterine fibroid) is a benign tumor that arise from the smooth muscle tissue of the uterus. They are common in women of reproductive age and can cause symptoms such as abdominal pain, pelvic pressure, and abnormal uterine bleeding. On physical examination, uterine fibroids may present as firm, smooth-surfaced masses located within the pelvis. Uterine fibroids are hormonally responsive tumors that can vary in size and location within the uterus. They are typically diagnosed based on clinical examination, imaging studies (such as pelvic ultrasound), and sometimes additional tests such as magnetic resonance imaging (MRI). Treatment options for uterine fibroids may include watchful waiting, medical management (e.g., hormonal therapy), or surgical intervention (e.g., myomectomy or hysterectomy) depending on the patient's symptoms, desire for fertility, and overall health status.

Ovarian tumors can present with abdominal pain and a palpable mass, they typically arise from the ovaries and may have a different characteristic appearance on examination. Ovarian tumors may be solid or cystic and are less likely to be centrally located within the pelvis.

Molar pregnancy, also known as gestational trophoblastic disease, typically present with vaginal bleeding, uterine enlargement, and elevated levels of human chorionic gonadotropin (hCG). While molar pregnancies can cause uterine enlargement, they are less likely to present as a palpable abdominopelvic mass.

Ectopic pregnancies occur when a fertilized egg implants outside of the uterus, most commonly within the fallopian tubes. While ectopic pregnancies can cause abdominal pain and may present as a palpable mass on examination, they are less likely to be centrally located within the pelvis.

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A 30-year-old gravida 1 para 0 woman arrives at the delivery room with pushing-down pain of 4 hours duration. During her antenatal care (ANC) follow-up, she was informed that she is carrying twins. Currently, her cervical dilation is 4 cm and the cervix is fully effaced. The fetal heart rate (FHR) of twin A is 130 bpm, and the FHR of twin B is 140 bpm. On ultrasound, twin A is in cephalic presentation, while twin B is in breech presentation. What is the best next step in the management of this patient's labor?

- ☐ Emergency cesarean section
- ☒ Follow labor progress
- ☐ Forceps delivery
- ☐ Attempt external cephalic version

Explanation :

The patient is in active labor with cervical dilation and effacement, and both fetuses have reassuring fetal heart rates. In the absence of any contraindications, it is appropriate to continue monitoring her labor progress and allow for vaginal delivery. The presentation of the second twin



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Explanation :

The patient is in active labor with cervical dilation and effacement, and both fetuses have reassuring fetal heart rates. In the absence of any contraindications, it is appropriate to continue monitoring her labor progress and allow for vaginal delivery. The presentation of the second twin (twin B) being breech is not an absolute indication for cesarean section in this scenario, especially if the first twin (twin A) is cephalic. The mode of delivery in twin pregnancies depends on various factors, including fetal presentation, gestational age, maternal health status, and obstetric considerations. In many cases, vaginal delivery is feasible and safe for twin pregnancies, especially when both fetuses are in favorable presentations and there are no contraindications to vaginal delivery. However, cesarean delivery may be indicated in certain situations, such as fetal distress, non-cephalic presentation of both twins, or obstetric complications. Forceps delivery may be considered in certain situations, such as prolonged second-stage labor or fetal distress, but it is not the preferred option in this case. Additionally, forceps delivery of the second twin in a twin delivery is rarely performed due to the increased risk of fetal injury. External cephalic version (ECV) is a maneuver used to attempt to turn a breech-presenting fetus to a cephalic presentation before labor begins. However, attempting ECV during active labor is not appropriate. Additionally, ECV would only be applicable for twin B, the breech-presenting twin, and not for twin A, who is already cephalic.

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A 36-year-old primigravida presents to the ED at 32 weeks' gestation with epigastric pain. Her BP is 150/100, but other vital signs are normal. While the nurse is performing his assessment in the room, the patient begins to seize. The next best step in management is:

- ☐ Hydralazine 10 mg IV push
- ☐ Lorazepam 2 mg/minute IV push
- ☐ Phenytoin 20 mg/kg IV
- ☒ Magnesium 6 g slow IV push

Explanation :

Eclampsia is a serious complication of pregnancy characterized by the onset of seizures (convulsions or fits) in a woman with preeclampsia. It is associated with hypertension, proteinuria, and end-organ dysfunction. Magnesium sulfate is the drug of choice for preventing and treating seizures in patients with eclampsia. It is administered intravenously as a loading dose followed by a maintenance infusion. Magnesium sulfate is effective in controlling seizures and reducing the risk of maternal morbidity and mortality associated with eclampsia. Seizures occurring within 24 hours after delivery are considered postpartum seizures. They may occur in the context of eclampsia



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Phenytoin is an antiepileptic medication used to prevent and treat seizures. However, it is not the first-line treatment for eclampsia or postpartum seizures. Phenytoin may be considered if magnesium sulfate is ineffective or contraindicated.

Diazepam is a benzodiazepine medication used to treat seizures. While it may be administered for acute seizure management in other settings, magnesium sulfate is the preferred agent for eclamptic seizures due to its specific effectiveness in preventing recurrent seizures and reducing the risk of maternal morbidity and mortality.

Hydralazine is an antihypertensive medication used to lower blood pressure in hypertensive emergencies. While controlling hypertension is important in the management of eclampsia, hydralazine is not indicated as the initial treatment for eclamptic seizures. Magnesium sulfate is the primary intervention to prevent and manage seizures in this context.

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A 28-year-old woman, para 4, who gave birth 2 months ago, presents with vaginal bleeding after sexual intercourse, which is her first time since delivery. On pelvic examination, there is a 6-week-sized uterus, and a fragile cervical mass is noted. What is the most likely diagnosis?

- ☐ Gestational trophoblastic disease
- ☒ Cervical cancer
- ☐ Uterine fibroid
- ☐ Cervical polyp

Explanation :

Cervical cancer is the fourth most common cancer in women worldwide. It is strongly associated with persistent infection with high-risk types of human papillomavirus (HPV). Risk factors for cervical cancer include early onset of sexual activity, multiple sexual partners, smoking, and



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GTD refers to a spectrum of conditions arising from abnormal proliferation of trophoblastic tissue. It typically presents with abnormal uterine bleeding, enlarged uterus, and elevated serum beta-hCG levels. However, the fragile cervical mass described in this case suggests a different pathology. Cervical polyps are benign growths that arise from the cervical canal or the endocervical mucosa. While they can cause postcoital bleeding, they are usually smooth and non-fragile. The description of a fragile cervical mass raises concern for a more serious pathology such as cervical cancer.

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A 28-year-old primigravida woman with a history of an uneventful pregnancy presents to the labor and delivery unit after being in active labor for 8 hours. On examination, her cervix is fully dilated (10 cm), with a cervical dilatation of 3, and 40% effaced. Fetal heart rate (FHR) monitoring reveals a heart rate of 110 beats per minute. On vaginal examination, the station is +2, and moderate caput and grade 2 molding are noted. What is the best next management step for this patient presenting in labor?

☐ Attempt forceps delivery due to prolonged labor

☒ Proceed with an emergency CS considering signs of fetal distress

☐ Augment labor with oxytocin to expedite delivery

☐ Continue labor observation with close fetal monitoring

Explanation :

Given the prolonged duration of labor, lack of progress, and signs of fetal distress (FHR of 110 bpm), an emergency CS is warranted to prevent potential complications to both the mother and the fetus. An emergency CS is performed when there is an immediate threat to the life or health of the mother or fetus. Common indications include fetal distress, cephalopelvic disproportion, placental abruption, cord prolapse, or failure to progress in labor. The decision to proceed with an emergency CS requires careful consideration of maternal and fetal well-being, and timely intervention can significantly reduce the risk of adverse outcomes.

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Forceps delivery may be considered in cases of prolonged second stage of labor or fetal distress, but in this scenario, fetal distress is evident with a heart rate of 110 bpm. Therefore, forceps delivery alone may not be the best initial choice. Augmentation of labor with oxytocin may be considered in cases of inadequate uterine contractions, but in this scenario, signs of fetal distress are present, indicating the need for more immediate intervention.

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A 2-day-old neonate presents with fast breathing and grunting for the past 10 hours. On physical examination, there is intercostal retraction and decreased air entry on auscultation. What is the most likely diagnosis in this newborn?

- ☒ Meconium aspiration syndrome (MAS)
- ☐ Neonatal sepsis
- ☐ Hyaline membrane disease (HMD)
- ☐ Pneumonia

Explanation :

Meconium aspiration syndrome (MAS) is when a newborn inhales meconium-stained amniotic fluid into the lungs, leading to airway obstruction, inflammation, and chemical pneumonitis. The classic clinical presentation includes respiratory distress, grunting, cyanosis, and signs of respiratory distress, such as intercostal retractions and decreased air entry. The history of fast breathing, grunting, and intercostal retractions in this case is consistent with MAS.

Neonatal sepsis presents with nonspecific signs such as fever, lethargy, poor feeding, and respiratory distress. While sepsis can cause respiratory symptoms, the presence of intercostal retractions and decreased air entry in this case suggests a primary pulmonary issue rather than sepsis.



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A 10-minute-old neonate born at 32wks was brought to the pediatric ICU with fast breathing, grunting, and cyanosis since birth. On physical examination, there is a bulged chest and decreased air entry on auscultation. What is the most likely diagnosis for this neonate?

- ☐ Pneumonia
- ☒ Hyaline membrane disease
- ☐ Neonatal sepsis
- ☐ Meconium aspiration syndrome

Explanation :

Hyaline membrane disease (HMD) also known as respiratory distress syndrome (RDS), is a common condition in premature infants due to surfactant deficiency. However, it can also occur in full-term infants, especially those born to diabetic mothers or delivered by cesarean section. The classic presentation includes rapid, shallow breathing, grunting, cyanosis, and chest wall retractions. The bulged chest and decreased air entry on auscultation are consistent with HMD.

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A 22-year-old primigravida presents to your community ED at 34 weeks of gestation with a chief complaint of headache and mild crampy abdominal pain. Her blood pressure is 160/100. Suspecting preeclampsia, you start a magnesium drip. While the patient is awaiting transfer 2 hours later, the nurse alerts you that she believes the patient is magnesium toxic. Which of the following is a sign of magnesium toxicity?

- ☐ Atrial fibrillation
- ☐ Increased deep tendon reflexes
- ☒ Somnolence
- ☐ Hyperventilation

Explanation :

Magnesium depresses the CNS and slows nerve conduction. It is used in preeclampsia to prevent progression to eclampsia, which is characterized by the presence of seizures. Magnesium slows neuromuscular conduction and decreases CNS irritability. Remembering this provides an easy means of remembering the actions of magnesium. It will decrease the respiratory rate, decrease deep tendon reflexes, and decrease the degree of consciousness. The loss of deep tendon reflexes is generally the first sign of magnesium toxicity.

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A 30-year-old woman delivered a healthy baby 12 hours ago. She had an uncomplicated vaginal delivery. Her prenatal course was largely uneventful except for two episodes of elevated blood pressure recorded during her last two prenatal visits, for which no treatment was initiated. She now presents with a sudden onset of generalized tonic-clonic seizures. On examination, her blood pressure is 170/110 mm Hg, and there is 3+ proteinuria on urinalysis. What is the most likely diagnosis, and what is the best initial management?

- ☒ Postpartum hemorrhage; administer intravenous oxytocin
- ☐ Eclampsia; administer intravenous magnesium sulfate



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☒ Postpartum hemorrhage; administer intravenous oxytocin

☐ Eclampsia; administer intravenous magnesium sulfate

☐ Hypoglycemia; administer intravenous dextrose

☐ Epilepsy; administer intravenous diazepam

Explanation :

Eclampsia is characterized by the onset of seizures in a woman with preeclampsia. Given the patient's recent delivery, episodes of elevated blood pressure during pregnancy, current high blood pressure, and proteinuria, eclampsia is the most likely diagnosis. The best initial management is to administer intravenous magnesium sulfate to control the seizures and prevent recurrence.

Postpartum hemorrhage is a common complication after delivery but does not typically present with seizures. The patient's elevated blood pressure and proteinuria point towards a hypertensive disorder rather than a hemorrhagic one. Administering oxytocin would help control bleeding but is irrelevant to seizure management.

Diazepam can be used to manage seizures, the patient's history and current presentation strongly suggest eclampsia, not epilepsy. Magnesium sulfate is preferred in eclampsia for seizure control and prevention.

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A 70-year-old man presents to the clinic with a 3-month history of abdominal swelling and fatigue. On examination, he has significant ascites and lower extremity edema. Laboratory tests show a serum albumin level of 3.0 g/dL and an ascitic fluid analysis reveals an albumin level of 1.7 g/dL. The serum-ascites albumin gradient (SAAG) is calculated to be 1.3 g/dL, and the ascitic protein is 2.0 g/dL. Which of the following is the most likely diagnosis?

☐ Tuberculous enteritis

☒ Cirrhosis

☐ Early Budd-Chiari syndrome

☐ Nephrotic syndrome

Explanation :

Cirrhosis is a common cause of ascites and typically presents with a high SAAG (≥ 1.1 g/dL) due to portal hypertension. The patient's SAAG of 1.3 g/dL supports this diagnosis. Additionally, the moderate ascitic protein level (2.0 g/dL) is consistent with cirrhotic ascites, which often has a lower protein content than transudative ascites from other causes.

Tuberculous enteritis can cause ascites, but it typically presents with a low SAAG (< 1.1 g/dL) because it is associated with peritoneal involvement leading to exudative ascites.

Budd-Chiari syndrome, which is caused by obstruction of the hepatic veins, can also lead to ascites with a high SAAG (≥ 1.1 g/dL). However, it often presents with additional symptoms such as hepatomegaly, severe abdominal pain, and rapid development of ascites, which are not described in this patient's history.

Nephrotic syndrome can cause ascites, but it typically presents with a low SAAG (< 1.1 g/dL).



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Nephrotic syndrome can cause ascites, but it typically presents with a low SAAG (< 1.1 g/dL) because the primary issue is hypoalbuminemia rather than portal hypertension.

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A 32-year-old woman, who is a para 1, presents to the clinic with concerns about infertility. She has been trying to conceive for the past 18 months without success. Her medical history is significant for a normal spontaneous vaginal delivery 4 years ago. She has regular menstrual cycles and no significant gynecological issues. A recent pregnancy test (HCG) was negative, and a transvaginal ultrasound showed a normal uterus and ovaries. What is the next best step in her evaluation?

- ☐ Hormone profile
- ☒ Hysterosalpingography
- ☐ Laparoscopy
- ☐ Endometrial biopsy

Explanation :

Hysterosalpingography (HSG) is a crucial investigation in the evaluation of infertility as it assesses the patency of the fallopian tubes and the uterine cavity. Since the patient has a history of a normal delivery, structural abnormalities or tubal blockages could have developed since then, making HSG the next best step to determine if there are any anatomical reasons for her infertility.

A hormone profile, including measurements of FSH, LH, estradiol, prolactin, and thyroid-stimulating hormone (TSH), is an important part of the infertility workup, especially to evaluate ovulatory function. However, given her regular menstrual cycles, it is likely that she is ovulating, so this may not be the most immediate next step compared to an HSG.

An endometrial biopsy is useful in diagnosing endometrial abnormalities, such as chronic endometritis or luteal phase defects, but it is not typically the first step in the evaluation of infertility, especially when there are no specific symptoms suggesting an endometrial pathology.



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A 30-year-old woman, gravida 1 para 1, presents to the clinic with amenorrhea for the past 3 months. She stopped using contraception 4 months ago in hopes of conceiving. Her menstrual cycles were regular before discontinuing contraception. She has had no periods since stopping contraception and reports no symptoms of pregnancy. A urine pregnancy test (HCG) was negative, and a recent transvaginal ultrasound showed a normal uterus and ovaries with no evidence of pregnancy or any structural abnormalities. What is the next best step in her evaluation?

☐ Serum HCG

☒ Progesterone challenge test

☐ Serum prolactin

☐ Thyroid function tests

Explanation :

A progesterone challenge test helps determine if the amenorrhea is due to anovulation or an estrogen deficiency. This is a reasonable next step for evaluating secondary amenorrhea in a non-pregnant woman with a normal ultrasound and negative urine HCG. If the patient bleeds after the test, it suggests she has sufficient estrogen but is not ovulating. If there is no bleeding, it may indicate a more complex hormonal issue or outflow tract obstruction.

Serum HCG test is more sensitive than a urine test, the urine HCG being negative and a normal ultrasound significantly reduce the likelihood of pregnancy. Therefore, further evaluation for other causes of amenorrhea should be considered before repeating the HCG test.

Elevated prolactin levels can cause amenorrhea and should be checked if there are symptoms of galactorrhea or other signs of hyperprolactinemia. However, it might not be the immediate next step without specific symptoms suggesting hyperprolactinemia.

Thyroid dysfunction can also cause menstrual irregularities, including amenorrhea. Thyroid function tests are an important part of the workup for secondary amenorrhea but given the typical presentation and recent cessation of contraception, a progesterone challenge test is usually considered first to evaluate ovarian function.

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A 20-year-old primigravida presents to the hospital in labor. During delivery, she experiences repeated convulsions. The healthcare team is unable to manage the convulsions effectively because essential medications are not available. Despite attempts to stabilize her, she deteriorates and ultimately dies during delivery. What is the most likely cause of her death?

☒ Delay in appropriate care

☐ Delay in referral

☐ Amniotic fluid embolism

☐ Cardiac arrest due to unknown cause

Explanation :

Delay in appropriate care is likely in this scenario because the patient experienced repeated convulsions (suggestive of eclampsia), and the healthcare team was unable to manage her condition due to the unavailability of essential medications. This indicates a failure in providing the necessary and timely medical intervention that could have prevented her death.

Delay in referral is less likely the primary cause in this context because the scenario does not indicate that there was an attempt to refer her to another facility. The issue was the lack of appropriate management at the current facility rather than a delay in transferring her to another one.

Amniotic fluid embolism is a rare but serious condition that can cause sudden cardiovascular



appropriate management at the current facility rather than a delay in transferring her to another one.

Amniotic fluid embolism is a rare but serious condition that can cause sudden cardiovascular collapse and death. However, the repeated convulsions point more towards eclampsia, a common complication in pregnancy associated with seizures, rather than an amniotic fluid embolism.

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A 24-year-old male with a history of Type 1 Diabetes Mellitus (T1DM) presents to the emergency department with symptoms of sweating, palpitations, shakiness, and confusion. He reports that he administered his regular morning dose of insulin but forgot to eat breakfast. His symptoms started approximately an hour after his insulin injection. On examination, his heart rate is 110 bpm, blood pressure is 90/60 mmHg, and he appears diaphoretic and anxious. What is the most likely diagnosis?

- ☐ Hyperglycemia
- ☐ Diabetic Ketoacidosis
- ☒ Hypoglycemia
- ☐ Hyperosmolar Hyperglycemic State

Explanation :

Hypoglycemia is the most likely diagnosis in this case. It occurs when blood glucose levels fall below normal. In Type 1 Diabetes Mellitus, insulin administration without adequate carbohydrate intake can cause blood glucose levels to drop too low, leading to symptoms such as sweating, palpitations, shakiness, confusion, and in severe cases, loss of consciousness. The patient's presentation of symptoms starting after skipping breakfast post-insulin administration aligns with hypoglycemia.

Hyperglycemia is characterized by high blood glucose levels, often presenting with symptoms like polyuria, polydipsia, and fatigue. This is unlikely in a patient who has administered insulin but did not eat, leading to a drop in blood glucose rather than an increase.

Diabetic Ketoacidosis (DKA) typically occurs due to a lack of insulin, leading to high blood glucose levels and ketone production. Symptoms include abdominal pain, vomiting, and deep, rapid breathing (Kussmaul respirations). This scenario does not fit the description of DKA as the patient administered insulin and his symptoms started after not eating, indicative of low rather than high blood glucose.

Hyperosmolar Hyperglycemic State (HHS) is a serious complication of diabetes characterized by extremely high blood glucose levels, severe dehydration, and altered consciousness. It primarily occurs in Type 2 Diabetes and does not fit the scenario of a patient with Type 1 Diabetes who administered insulin but forgot to eat.

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A 2-year-old child is brought to the clinic by his mother due to oozing skin lesions that have been present for the past 2 months. The lesions are primarily located on the face, scalp, and extensor surfaces of the arms and legs. The child is frequently scratching the affected areas, and there are signs of excoriation. The mother has a history of asthma and notes that the child has had similar episodes in the past, which tend to worsen with exposure to certain foods and environmental allergens. What is the most likely diagnosis?

- ☐ Seborrheic dermatitis
- ☒ Atopic dermatitis
- ☐ Contact dermatitis
- ☐ Psoriasis



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- ☒ Atopic dermatitis
- ☐ Contact dermatitis
- ☐ Psoriasis

Explanation :

Atopic dermatitis (AD) is a chronic inflammatory skin condition that commonly presents in early childhood. It is characterized by pruritic (itchy) lesions, which can become oozing and crusted due to scratching. Lesions are typically found on the face, scalp, and extensor surfaces in infants and young children. The presence of a family history of atopy (e.g., the mother's asthma) supports this diagnosis. The chronic nature of the lesions and the exacerbation by environmental factors are also characteristic of atopic dermatitis.

Seborrheic dermatitis typically presents as greasy, scaly, erythematous plaques on areas rich in sebaceous glands such as the scalp (cradle cap in infants), face, and trunk. It usually doesn't cause intense itching or excoriations and is not typically associated with a family history of atopy (e.g., asthma, allergic rhinitis, or atopic dermatitis).

Contact dermatitis results from skin exposure to allergens or irritants, leading to localized rash and itching. However, it is typically confined to the area of contact and does not usually present as a chronic condition with widespread involvement as described in this case.

Psoriasis in children can present with well-demarcated, thick, scaly plaques but it usually does not have the oozing or intense itching seen in atopic dermatitis. It is also less commonly seen in the facial and extensor distribution typical of atopic dermatitis in young children.

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A 15-year-old girl presents to the dermatology clinic with concerns about acne. She reports that she has had acne for about a year, but it has recently become more noticeable. On physical examination, there are multiple small, flesh-colored to slightly dark lesions on her forehead, nose, and chin. The lesions are not inflamed and there is no erythema or pus. She denies any pain or significant itching. The patient is otherwise healthy and does not take any medications. What is the most likely type of acne lesion she has?

- ☐ Papules
- ☐ Pustules
- ☐ Nodules

☒ Comedones

Explanation :

Comedones are the hallmark of non-inflammatory acne and include both open comedones (blackheads) and closed comedones (whiteheads). They are caused by the blockage of hair follicles by sebum and keratin. Open comedones appear as small, dark spots, while closed comedones are small, flesh-colored bumps. The description of small, flesh-colored to slightly dark lesions without inflammation fits the presentation of comedones. The treatment for comedonal



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Comedones are the hallmark of non-inflammatory acne and include both open comedones (blackheads) and closed comedones (whiteheads). They are caused by the blockage of hair follicles by sebum and keratin. Open comedones appear as small, dark spots, while closed comedones are small, flesh-colored bumps. The description of small, flesh-colored to slightly dark lesions without inflammation fits the presentation of comedones. The treatment for comedonal acne typically includes topical retinoids, which help to normalize the shedding of dead skin cells and reduce the formation of new comedones. Other treatments may include benzoyl peroxide, salicylic acid, or alpha hydroxy acids. Proper skin hygiene and avoiding comedogenic products are also important.

Papules are small, raised, solid pimples or swellings, often part of a rash, that typically do not contain pus. They are usually inflamed and may be red or pink. This description does not fit the patient's non-inflamed, flesh-colored lesions.

Pustules are small, inflamed, pus-filled, blister-like sores (pimples). They are a more severe form of acne lesion characterized by infection and inflammation. The patient's lesions are non-inflamed and do not contain pus, making pustules an unlikely diagnosis.

Nodules are large, painful, solid pimples or lumps that lie deep within the skin. They are a severe form of acne and can cause significant scarring. The patient's description of mild, non-inflamed lesions does not match the characteristics of nodules.

67

A 23-year-old medical student presents to the clinic with a 6-month history of recurrent abdominal pain and watery diarrhea. The pain typically worsens before exams or stressful events and is relieved by defecation. He reports that he does not wake up at night due to these symptoms. He denies any blood in his stool, weight loss, or fever. His past medical history is unremarkable, and he is not on any medications. Physical examination is unremarkable. What is the most likely diagnosis?

- ☒ Irritable Bowel Syndrome
- ☐ Psychogenic Diarrhea
- ☐ Inflammatory Bowel Disease
- ☐ Celiac Disease

Explanation :

Irritable Bowel Syndrome (IBS) is a functional gastrointestinal disorder characterized by chronic abdominal pain and altered bowel habits without any organic cause. The patient's symptoms of recurrent abdominal pain, watery diarrhea, and symptom exacerbation with stress, along with relief of pain after defecation, are classic features of IBS. The absence of alarm symptoms such as



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While stress can exacerbate symptoms in patients with IBS, psychogenic diarrhea typically refers to diarrhea purely of psychological origin without accompanying chronic abdominal pain or specific patterns associated with bowel movements. The presence of recurrent abdominal pain that is relieved by defecation aligns more closely with IBS. Treatment typically involves dietary modifications (such as a low FODMAP diet), stress management, and medications to alleviate specific symptoms (antispasmodics, laxatives, or antidiarrheals).

68

A 37-year-old male with a history of chronic alcohol use presents to the clinic with a 3-month history of dyspeptic symptoms, including epigastric pain, bloating, and occasional nausea. He denies any significant weight loss or changes in bowel habits. On physical examination, he appears pale. Laboratory results reveal a hemoglobin level of 6.7 g/dL. The patient has no prior history of gastrointestinal bleeding or similar symptoms. What is the most alarming reason for referral to a gastroenterologist?

Anemia

- ☐ Age
- ☐ Alcoholic history
- ☐ Dyspeptic symptoms

Explanation :

The patient's hemoglobin level of 6.7 g/dL is significantly low, indicating severe anemia. This level of anemia is alarming and warrants urgent evaluation by a gastroenterologist to identify the underlying cause, which could be gastrointestinal bleeding, chronic gastritis, or another serious condition. Anemia at this severity could also have significant systemic effects and requires prompt investigation and management.

69

A 35-year-old woman, para 5, presents to the clinic with complaints of mood swings and loss of libido for the past month. She reports that these symptoms have been affecting her relationship with her husband. She has been experiencing these symptoms for the past 10 months. She has no significant



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69

A 35-year-old woman, para 5, presents to the clinic with complaints of mood swings and loss of libido for the past month. She reports that these symptoms have been affecting her relationship with her husband. She has been using contraception for the past 10 months. She has no significant past psychiatric history and no other medical conditions. Physical examination and routine blood tests are unremarkable. What is the most likely diagnosis?

☐ Perimenopause☒ Contraception side effect☐ Psychiatric disorder☐ Relationship issues**Explanation :**

Hormonal contraceptives can cause side effects such as mood swings, loss of libido, and other psychological symptoms. Given the timeline of symptom onset 10 months after starting contraception, this is the most likely diagnosis. Different types of hormonal contraceptives can have different side effects, and it's not uncommon for women to experience these issues. Perimenopause typically starts in women in their 40s and 50s, leading up to menopause. While mood swings and loss of libido can be symptoms of perimenopause, this diagnosis is less likely given the patient's age of 35. It is possible but not the most likely cause in this scenario.

70

A 50-year-old male presents to the emergency department with a 24-hour history of abdominal pain and vomiting. He describes the pain as central, crampy, and intermittent. On examination, his abdomen is distended but soft, with hyperactive bowel sounds. There is no tenderness, guarding, or rigidity. His vital signs are stable, and he has no fever. What is the most likely diagnosis?

☒ Simple small bowel obstruction☐ Strangulated small bowel obstruction☐ Gangrenous bowel obstruction☐ Large bowel obstruction**Explanation :**

Simple small bowel obstruction is the most likely diagnosis given the patient's symptoms and physical examination findings. Simple small bowel obstruction presents with central, crampy abdominal pain, vomiting, abdominal distension, and hyperactive bowel sounds. The absence of peritoneal signs (no tenderness, guarding, or rigidity) suggests that there is no ischemia or perforation, making it a simple obstruction.

Strangulated small bowel obstruction occurs when the blood supply to a segment of the bowel is compromised. It typically presents with more severe, continuous pain, signs of peritonitis (tenderness, guarding, rigidity), fever, and signs of systemic toxicity. The patient's stable vital signs and lack of tenderness make strangulation less likely.

Gangrene of the bowel is a severe complication of strangulated bowel obstruction, leading to necrosis of the bowel wall. It presents with severe abdominal pain, signs of sepsis (fever, tachycardia, hypotension), and peritoneal signs. The absence of these severe symptoms and peritoneal signs makes gangrenous obstruction unlikely in this patient.

Large bowel obstruction can present with similar symptoms, it is typically associated with gradual onset, lower abdominal pain, and less vomiting compared to small bowel obstruction. The clinical features described (central, crampy pain, and hyperactive bowel sounds) are more characteristic of a small bowel obstruction.



71

Which of the following are the two most common causes of diarrhea in AIDS patients?

- ☒ Cryptosporidium and cytomegalovirus
- ☐ C. difficile and Salmonella
- ☐ Giardia and Mycobacterium avium-intracellulare
- ☐ Giardia and Isospora

Explanation :

CMV causes one-fifth to one-third of all cases of diarrhea in AIDS patients.

72

A 3-year-old male child presents with his mother, who reports that she noticed the absence of one testicle in the scrotum one month ago. On physical examination, the testicle is palpated in the inguinal canal. The child underwent surgical correction (orchiopexy) to reposition the testicle into the scrotum. Despite surgical correction, what risk remains high for this child?

- ☐ Infertility
- ☒ Malignancy
- ☐ Torsion
- ☐ Hernia recurrence

Explanation :

The risk of testicular cancer remains elevated even after orchiopexy. Cryptorchidism is a known risk factor for testicular cancer, and although surgical correction helps, it does not eliminate the increased risk. Testicular cancer risk remains higher in individuals with a history of undescended testicles compared to the general population. The condition of undescended testicles is associated with increased risks of infertility and malignancy. Early surgical intervention (orchiopexy) can mitigate some of these risks, especially if performed before the child reaches 1 year of age. The lifetime risk of developing testicular cancer is higher in individuals with a history of cryptorchidism, making regular testicular self-examination and periodic follow-up important for early detection. Infertility is a concern for undescended testicles (cryptorchidism), the risk is significantly reduced after early surgical correction, especially if the surgery is performed before 1-2 years of age. Since this child is already 3 years old, there might be some increased risk, but it is not as high as the risk of malignancy.

Testicular torsion is a potential complication of undescended testicles; however, orchiopexy reduces this risk significantly as the testicle is fixed within the scrotum.

Inguinal hernias can be associated with undescended testicles, the primary concern after orchiopexy is not the recurrence of hernias but the long-term risk of malignancy and, to a lesser extent, infertility.

73

The classic sequence of color changes in the fingers of patients experiencing Raynaud phenomenon is:

- ☐ Blue to white to red
- ☐ Red to white to blue



73

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- ☐ Blue to white to red
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- ☒ White to blue to red

Explanation :

Raynaud phenomenon is nearly universal in patients with scleroderma (also known as systemic sclerosis) and is the earliest sign of the disease. Raynaud phenomenon is also common in patients with lupus and RA. The exact mechanism of Raynaud phenomenon in scleroderma (or in other autoimmune diseases) is not known. Raynaud phenomenon occurs in three phases. Initially, digital pallor (white) results from total closure of the palmar and digital arteries causing a cessation in digital blood flow. When mild relaxation occurs, a trickle of blood is able to perfuse the ischemic digit but the hemoglobin is rapidly desaturated resulting in cyanosis (blue). Finally, arterial spasm resolves and restores blood flow to baseline, resulting in a reactive hyperemia (red).

74

A 40-year-old male presents to the emergency department with a painful swelling that extends from his submandibular area to his clavicle. The swelling is tender but non-fluctuant. He reports difficulty swallowing and a muffled voice. His vital signs are notable for a fever of 38.5°C. Physical examination reveals induration and tenderness of the submandibular region without fluctuance. There is also noticeable trismus (difficulty opening the mouth) and drooling. What is the most likely diagnosis?

- ☐ Retropharyngeal abscess
- ☒ Ludwig's angina
- ☐ Parotid gland abscess
- ☐ Cervical lymphadenitis

Explanation :

Ludwig's angina is a serious, life-threatening cellulitis of the submandibular space that often results from a dental infection. It presents with rapidly progressing, bilateral, submandibular, and sublingual swelling. Key features include induration (non-fluctuant swelling), pain, trismus, dysphagia (difficulty swallowing), and a "woody" feeling of the submandibular area. The swelling can extend down to the clavicle, and the patient may have systemic signs of infection such as fever. Immediate management includes securing the airway, intravenous antibiotics, and often surgical drainage. CT imaging of the neck can help confirm the diagnosis and assess the extent of the infection. If untreated, Ludwig's angina can lead to airway obstruction, sepsis, and mediastinitis, making early recognition and treatment critical.

Retropharyngeal abscess typically presents with neck stiffness, difficulty swallowing, and fever. However, the swelling is usually more posterior and does not typically extend from the submandibular area to the clavicle in the described manner. Retropharyngeal abscesses are also more commonly seen in children rather than adults.

Parotid gland abscess would typically present with swelling over the parotid gland area (near the ear and angle of the jaw), and it is often associated with purulent discharge from the parotid duct, pain, and fever. The described distribution of swelling extending to the clavicle is not consistent with a parotid gland abscess.

Cervical lymphadenitis typically presents with enlarged cervical lymph nodes, which can be tender and swollen.



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Cervical lymphadenitis involves inflamed cervical lymph nodes, which can be tender and swollen. However, the distribution from the submandibular region to the clavicle suggests a more extensive process. Lymphadenitis would also typically present with discrete nodules rather than a continuous, non-fluctuant swelling.

75

A 25-year-old athletic male presents to the clinic with a scrotal swelling that he noticed after a heavy weightlifting session. He reports that the swelling is not painful but has been persistent. On physical examination, the swelling increases with standing and straining. When the deep inguinal ring is occluded (Valsalva maneuver or coughing), the bulge does not disappear. The swelling is located in the scrotum, and there is no associated redness or tenderness. What is the most likely type of hernia?

- ☐ Indirect inguinal hernia
- ☒ Direct inguinal hernia
- ☐ Femoral hernia
- ☐ Hydrocele

Explanation :

Direct inguinal hernias protrude through the posterior wall of the inguinal canal, medial to the inferior epigastric vessels, and are not affected by occlusion of the deep inguinal ring. These hernias typically occur in the region of the Hesselbach triangle and are more common in older men but can occur in younger individuals, especially those involved in strenuous activities like heavy lifting. The treatment for symptomatic inguinal hernias typically involves surgical repair to prevent complications such as incarceration or strangulation of the herniated tissue. Conservative management may be considered for asymptomatic or minimally symptomatic cases, especially in older adults with significant comorbidities.

Indirect inguinal hernias pass through the deep inguinal ring and can extend into the scrotum. When the deep inguinal ring is occluded, the bulge typically reduces or disappears. The patient's swelling did not change with occlusion of the deep inguinal ring, making this diagnosis less likely.

Femoral hernias appear below the inguinal ligament and are more common in women. They often present as a bulge in the upper thigh or groin and are not associated with scrotal swelling.

A hydrocele is a fluid-filled sac surrounding a testicle that causes scrotal swelling. It is usually non-tender and does not change with straining or occlusion of the inguinal ring. Transillumination of the scrotum can help differentiate it from a hernia, as hydroceles will typically allow light to pass through.



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An 11-month-old infant presents to the emergency department with a history of diarrhea for the past 2 days. The child is now lethargic and has significantly reduced oral intake over the past 24 hours. The infant weighs 15 kg. On physical examination, the child has signs of dehydration including dry mucous membranes, sunken eyes, and decreased skin turgor. The heart rate is elevated, and capillary refill time is prolonged. What is the most appropriate management for this child?

- ☐ Administer 100 ml/kg of normal saline over 5 hours
- ☐ Administer 75 ml/kg of ORS over 4 hours
- ☒ Administer 20 ml/kg of NS bolus and reassess
- ☐ Continue breastfeeding and give additional fluids ad libitum

Explanation :

For infants with severe dehydration, the initial step is to rapidly restore intravascular volume. A bolus of 20 ml/kg of NS (300 ml for this child) is appropriate, followed by reassessment to determine if additional boluses are needed.

Oral rehydration is appropriate for mild to moderate dehydration, but this child is lethargic and unable to feed, indicating severe dehydration requiring IV fluids.

The standard approach for severe dehydration includes administering a 20 ml/kg bolus of isotonic fluid (NS or lactated Ringer's solution). This can be repeated if necessary, based on the clinical response. After each bolus, the child should be reassessed for improvements in mental status, heart rate, capillary refill time, and overall hydration status. Additional boluses may be required if there is no significant improvement.

77

Which of the following muscles is most commonly injured in rotator cuff tears?

- ☒ Supraspinatus
- ☐ Infraspinatus
- ☐ Subscapularis
- ☐ Teres minor

Explanation :

The rotator cuff is a group of muscles composed of the supraspinatus, infraspinatus, teres minor, and subscapularis (SITS). Its primary function is to stabilize the shoulder joint. It is most commonly injured chronically with repetitive motions or acutely with a fall on an outstretched hand. The supraspinatus is the most common of the rotator cuff muscles to be injured. Physical examination of patients with supraspinatus tears indicates inability to maintain active abduction at 90 degrees without limitation in passive range of motion. Diagnosis may be confirmed with MRI and acute tears may require early surgery. The deltoid muscle is not technically part of the rotator cuff.

78

A referral hospital is experiencing significant noise pollution in its operation room due to an old and noisy ceiling fan. The constant noise is disrupting the doctors' ability to concentrate and affecting their work environment. The hospital administration is considering several options to address this problem and improve the OR conditions for the doctors and nurses. What is the most appropriate action to solve this problem?



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- ☒ Buy a new, quieter fan
- ☐ Make the walls soundproof
- ☐ Replace the fan with air conditioning
- ☐ Move the students to another classroom

Explanation :

Buy a new, quieter fan is the most appropriate and cost-effective solution. Replacing the old fan with a new, quieter model directly addresses the source of the noise pollution. This solution is likely to be less disruptive and less expensive than other options, and it can be implemented relatively quickly.

79

In a hospital setting, two groups of hospital staff are having an argument over the use of a shared room. The disagreement escalates, but ultimately, one of the groups decides to leave the shared room and move to another room to resolve the conflict. What type of conflict resolution strategy is demonstrated in this scenario?

- ☒ Accommodation
- ☐ Avoidance
- ☐ Collaboration
- ☐ Competition

Explanation :

Accommodation is a conflict resolution strategy where one party gives in to the wishes or demands of another. In this scenario, one group decides to leave the shared room and move to another room, thereby accommodating the other group's desire to use the shared space. This action helps to resolve the conflict by sacrificing their own needs for the benefit of the other group.

Avoidance involves ignoring the conflict or withdrawing from the situation without addressing the issue directly. In this scenario, the group moving to another room is making an active decision to resolve the conflict, which does not align with avoidance.

Collaboration involves both parties working together to find a mutually beneficial solution. This scenario does not describe a collaborative effort, as only one group made a concession without any indication of joint problem-solving or mutual agreement.

Competition involves one party pursuing their own interests at the expense of the other party. This scenario does not fit the competitive strategy, as there is no indication of one group aggressively pursuing their own needs over the other group's needs.

80

A female patient presents with right upper quadrant colicky pain. Ultrasound imaging reveals a 10 mm dilated common bile duct and a hyperechoic mass. What is the most appropriate management strategy for this patient?



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- ☒ Emergency surgery
- ☐ Start on vitamin K for 3 days and surgery after that
- ☐ Appointment after 6 weeks
- ☐ Conservative management with pain relief medications

Explanation :

Given the presence of right upper quadrant colicky pain, a dilated common bile duct, and a hyperechoic mass on ultrasound, this suggests the possibility of a common bile duct stone causing obstruction. In such cases, prompt intervention with emergency surgery, such as laparoscopic cholecystectomy with common bile duct exploration, is warranted to alleviate the obstruction and prevent complications like cholangitis or pancreatitis.

81

A 7-year-old child presents with verrucous lesions on the back of her hand and the tip of her finger. The lesions are rough, raised, and have a cauliflower-like appearance. The child is otherwise healthy, with no other significant medical history. What is the most likely diagnosis?

- ☒ Common wart (Verruca vulgaris)
- ☐ Molluscum contagiosum
- ☐ Flat warts (Verruca plana)
- ☐ Seborrheic keratosis

Explanation :

Common wart (Verruca vulgaris) are caused by human papillomavirus (HPV) and typically present as rough, raised lesions with a verrucous (cauliflower-like) appearance. They are frequently found on the hands and fingers, which aligns with the described presentation. Treatment options for common warts include cryotherapy, salicylic acid preparations, and various other topical treatments. Many warts resolve spontaneously over time, particularly in children.

Molluscum contagiosum presents as smooth, dome-shaped, pearly lesions with a central umbilication. It is caused by a poxvirus and is more commonly seen on the trunk and extremities, rather than as verrucous lesions on the hands and fingers.

Flat warts (Verruca plana) are also caused by HPV, but they are typically smooth, flat-topped, and less raised compared to common warts. They are usually found on the face, neck, and back of the hands but do not have the rough, cauliflower-like appearance described in the case.

Seborrheic keratosis typically appears as well-demarcated, waxy, stuck-on lesions that are more common in older adults. They do not have the verrucous, cauliflower-like appearance of common warts and are uncommon in children.

82

A 19-year-old female presents with lower abdominal pain and cervical motion tenderness on examination. She reports having been treated for gonorrhea 4 months ago and has had a new sexual partner for the past 3 months. Given her history and current presentation, what is least likely to benefit her? What is the least likely to benefit her management at this time?



82

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- ☐ Test for sexually transmitted infections
- ☒ Test for cervical cancer
- ☐ Empiric antibiotic treatment for pelvic inflammatory disease (PID)
- ☐ Counseling and education on safe sex practices

Explanation :

Screening for cervical cancer is important, it is unlikely to benefit her immediate management in the context of acute pelvic pain and suspected PID. Cervical cancer screening is generally recommended starting at age 21, and her current symptoms are more indicative of an infectious process rather than malignancy.

83

A recent study investigated the relationship between family history of diabetes mellitus (DM) and the likelihood of developing DM. The study found an odds ratio (OR) of 1.5 for individuals with a family history of DM. What does an odds ratio of 1.5 for those with a family history of DM indicate?

- ☐ 1.5 times more likely for DM patients to report family history
- ☐ 150% more likely for DM patients to report family history
- ☒ 1.5 times more likely for those with family history to have DM
- ☐ 150% more likely for those with family history to have DM

Explanation :

1.5 times more likely for those with family history to have DM is the correct interpretation of an odds ratio. An odds ratio of 1.5 means that individuals with a family history of DM are 1.5 times more likely to develop DM compared to those without a family history. Odds Ratio (OR) is a measure of association between an exposure and an outcome. An OR of 1.0 indicates no association, greater than 1.0 indicates an increased likelihood of the outcome given the exposure, and less than 1.0 indicates a decreased likelihood. An OR of 1.5 means the odds of the outcome are 1.5 times (or 50% higher) in the exposed group compared to the unexposed group. It does not mean 150% higher, which would correspond to an OR of 2.5 (1 + 1.5).

84

A 28-year-old female presents to the emergency department with severe lower abdominal pain and vaginal bleeding. She has missed her last two menstrual periods. On examination, her abdomen is tender with rebound tenderness. A transvaginal ultrasound reveals an empty uterine cavity and a mass in the lower abdomen suggestive of an abdominal ectopic pregnancy. Her vital signs are as follows: BP 90/60 mmHg, HR 120 bpm, RR 22 breaths per minute. Her hemoglobin is 8 g/dL. What is the most appropriate next step in the management of this patient?

- ☐ Observation and repeat ultrasound in 24 hours
- ☐ Methotrexate administration



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- ☐ Observation and repeat ultrasound in 24 hours
- ☐ Methotrexate administration
- ☒ Emergency laparotomy
- ☐ Expectant management with serial beta-hCG levels

Explanation :

The patient has signs of hemodynamic instability (low blood pressure, high heart rate) and clinical features suggesting an abdominal ectopic pregnancy, which can cause significant internal bleeding. An emergency laparotomy is necessary to control the bleeding and remove the ectopic pregnancy. The management of an ectopic pregnancy depends on the location and the patient's stability. For hemodynamically unstable patients or those with significant symptoms, surgical intervention is required. Methotrexate can be considered for stable patients with early, unruptured ectopic pregnancies. Expectant management is rarely used and only in very specific, stable cases.

85

A 45-year-old male presents to the emergency department with a sudden onset of severe headache, described as "the worst headache of my life." The headache began abruptly 2 hours ago while he was watching TV and was accompanied by nausea and vomiting. He has a history of hypertension but is otherwise healthy. On physical examination, he appears distressed, with photophobia and a stiff neck. His vital signs are BP 160/100 mmHg, HR 90 bpm, RR 18 breaths per minute, and temperature 37.2°C (99°F). A non-contrast CT scan of the head is performed. What is the most likely diagnosis?

- ☐ Migraine
- ☒ Subarachnoid hemorrhage
- ☐ Tension headache
- ☐ Meningitis

Explanation :

Subarachnoid hemorrhage: A sudden, severe headache described as "the worst headache of my life" is classic for a subarachnoid hemorrhage. The presence of nausea, vomiting, photophobia, a stiff neck, and hypertension supports this diagnosis. Subarachnoid hemorrhage is a medical emergency typically caused by the rupture of an aneurysm or arteriovenous malformation. It often presents with a sudden, severe headache, known as a "thunderclap headache," and may be associated with neck stiffness, nausea, vomiting, and altered mental status. A non-contrast CT scan of the head is the initial imaging study of choice to confirm the diagnosis. If the CT scan is negative but suspicion remains high, a lumbar puncture may be performed to look for xanthochromia. Early diagnosis and management are crucial to prevent complications and improve outcomes. Migraines can cause severe headaches, they typically have a gradual onset and are often preceded by an aura. Migraines also commonly present with throbbing pain and may have triggers, which are not described in this case. The description of the headache as "the worst headache of my life" and the abrupt onset are not typical of migraines.

Tension headaches are usually characterized by a dull, pressing pain that is mild to moderate in



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Tension headaches are usually characterized by a dull, pressing pain that is mild to moderate in intensity. They do not present with sudden onset or severe symptoms such as photophobia and a stiff neck. Therefore, a tension headache is unlikely in this scenario.

86

A patient enrolled in a research study that was originally promised to take six months to complete. However, the study extended beyond the initial timeframe, and during this period, the patient developed other health issues. The research team treated these additional conditions at no cost to the patient. Which ethical principle best describes the research team's decision to treat the patient's additional health issues for free?

- ☒ Beneficence
- ☐ Justice
- ☐ Non-maleficence
- ☐ Autonomy

Explanation :

Beneficence refers to the ethical principle of doing good and acting in the best interest of the patient. By treating the patient's additional health issues for free, the research team is ensuring the well-being of the patient and acting to benefit them, which aligns with the principle of beneficence. Justice refers to fairness and equality in the distribution of resources and treatment. While treating the patient for free could be seen as fair, justice is more about ensuring that all participants have equal access to care and resources, rather than focusing on the benefit to an individual patient. Non-maleficence is the principle of "do no harm." While this principle ensures that the research does not cause harm to the patient, the decision to treat the patient's additional conditions for free is more aligned with providing benefit rather than simply avoiding harm. Autonomy refers to respecting the patient's right to make informed decisions about their own care. This principle is not directly related to the research team's decision to provide free treatment for additional health issues.



87

A 30-year-old male patient was counseled on HIV testing and subsequently tested positive for HIV. He requested that the nurse not disclose his HIV status to his family. The nurse is now considering how to handle the situation. What should the nurse do?

☒ Respect his decision and maintain confidentiality

☐ Tell his family

☐ Notify public health authorities

☐ Advise him to inform his family

Explanation :

Healthcare professionals are obligated to maintain patient confidentiality and respect their wishes unless there is a clear and immediate risk to others. In this case, the patient's HIV status is confidential information that should not be disclosed without his consent. Healthcare providers are ethically and legally obligated to protect patient information. Disclosing a patient's HIV status without consent can have serious personal, social, and legal repercussions for the patient.

88

A manager at a healthcare facility believes that his subordinates will not work properly if left to their own devices. As a result, he makes all the decisions himself and closely supervises all tasks. What type of management style is the manager employing?

☒ Autocratic

☐ Democratic

☐ Laissez-faire

☐ Transformational

Explanation :

An autocratic management style involves a manager making all the decisions and closely supervising and controlling employees. This style is characterized by individual control over all decisions with little input from group members. It is effective in situations where quick decision-making is crucial, but it can also lead to low employee morale and creativity over time.

Democratic management style involves the manager including one or more employees in the decision-making process (determining what to do and how to do it). However, the manager maintains the final decision-making authority. It is characterized by collaboration and shared decision-making, which is not described in this scenario.

Laissez-faire style is characterized by a hands-off approach where the manager provides little or no direction and allows employees to make decisions. It fosters creativity and independence but can lead to a lack of direction and accountability. This is the opposite of what is described in the scenario.

Transformational style involves managers who work with their teams to identify needed change, creating a vision to guide the change through inspiration, and executing the change in tandem with committed members of the group. This style focuses on leadership and motivation, which is not evident in the scenario provided.

89

A 60-year-old female is seen 7th post-operative day after thyroid surgery. She reports experiencing paresthesia and muscle cramps. On examination, she has a positive Chvostek sign and Trousseau's sign. What is the most likely electrolyte abnormality?



89

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- ☒ Hypocalcemia
- ☐ Hypercalcemia
- ☐ Hyponatremia
- ☐ Hyperkalemia

Explanation :

Hypocalcemia Paresthesia (tingling sensation) and muscle cramps are classic symptoms of hypocalcemia. The positive Chvostek and Trousseau signs are specific physical examination findings that indicate low calcium levels. Post-thyroidectomy, hypocalcemia often occurs due to inadvertent removal or damage to the parathyroid glands, which are responsible for calcium regulation.

- Chvostek Sign is elicited by tapping the facial nerve in front of the ear, which causes twitching of the facial muscles if hypocalcemia is present.
- Trousseau Sign is elicited by inflating a blood pressure cuff above systolic pressure for several minutes, which causes carpal spasm if hypocalcemia is present.

90

A 35-year-old female patient undergoes epigastric hernia repair. During the procedure, she repeatedly complains of pain and a dragging sensation. The surgical team administers a local anesthetic, and within 5 minutes, she experiences paresthesia, dizziness, and tinnitus. What is the most likely cause of her symptoms?

- ☐ Hypoglycemia
- ☐ The adrenaline
- ☒ Lidocaine toxicity
- ☐ Anxiety

Explanation :

Lidocaine toxicity can occur if the drug is administered in too high a dose or inadvertently into a blood vessel. Symptoms of lidocaine toxicity include paresthesia, dizziness, tinnitus, confusion, and even seizures. The rapid onset of these neurological symptoms after local anesthetic administration strongly suggests lidocaine toxicity. It occurs when the blood levels of lidocaine become too high, which can happen due to overdose or inadvertent intravascular injection. Early symptoms include circumoral numbness, tinnitus, and dizziness. Severe toxicity can lead to seizures, cardiac arrhythmias, and cardiac arrest.

Immediate steps in management include stopping the administration of lidocaine, ensuring airway protection, providing oxygen, and potentially administering intravenous lipid emulsion therapy in severe cases.

91

Which of the following is the most common complication of Epstein-Barr virus (EBV) pharyngitis?

- ☒ Asymptomatic elevated transaminases

Airway obstruction



91

Which of the following is the most common complication of Epstein-Barr virus (EBV) pharyngitis?

☒ Asymptomatic elevated transaminases

☐ Airway obstruction

☐ Splenic rupture

☐ Hemolytic anemia

Explanation :

Elevations of the hepatic enzymes AST and ALT commonly occur (up to 80%) to approximately four times the upper limit of normal and peak during the second to fourth week of illness. However, levels well over 1,000 may occur. Jaundice may also be seen (5%), but coagulopathy is not typical. Spontaneous recovery is the rule, and supportive care is the only required treatment. Airway obstruction is probably the most important complication for the emergency medicine physician. It is due to tonsillar hypertrophy and occurs more commonly in young children because of their prominent lymphoid tissue. Overall, however, this complication occurs in < 5% of patients and can usually be managed conservatively by elevating the head of the bed, giving intravenous fluids and corticosteroids and using humidified air. Splenic rupture receives a lot of attention but occurs in < 0.5% of adults with EBV infection. The rate in children is thought to be much lower. It most commonly occurs in the second and third week of illness and is rarely fatal when it does occur. Patients present with left upper quadrant abdominal pain, with or without radiation to the left shoulder (Kehr sign), and splenic rupture should be considered especially in the setting of shock. Numerous hematologic complications are associated with EBV infection, including autoimmune hemolytic anemia (3%), mild thrombocytopenia (25% to 50%), and mild transient neutropenia (50% to 80%). Headache is the most common neurologic manifestation of EBV infection (50%), although meningoencephalitis is the most common severe neurologic complication (1% to 5%).

92

A 55-year-old female presents with 1 to 2 days of palpitations, anxiety, agitation, tachycardia, and hypertension. Her thyroid-stimulating hormone level is very low and her free T4 and free T3 levels are elevated. Which of the following is the most appropriate sequence in which to administer ideal therapies?

☐ Potassium iodide → Propranolol → Propylthiouracil → Dexamethasone

☐ Potassium iodide → Propylthiouracil → Dexamethasone → Propranolol

☐ Dexamethasone → Propranolol → Propylthiouracil → Potassium iodide

☒ Propylthiouracil → Potassium iodide → Propranolol → Dexamethasone

Explanation :

Thyroid storm treatment requires sequential inhibition of the following: hormone synthesis, hormone release, and peripheral conversion from T4 to T3. Thionamides, including propylthiouracil and methimazole, prevent the synthesis of new thyroid hormone. Potassium iodide serves as a source of inorganic iodide, which prevents release of preformed hormone from the thyroid gland. In the acute setting potassium iodide can actually promote new hormone synthesis and should, therefore, always be given after thionamide therapy in thyroid storm. Propranolol is used next to treat cardiovascular effects of thyrotoxicosis as well as to block peripheral conversion of T4 to the more metabolically active T3. Glucocorticoids are used to treat any underlying adrenal insufficiency and may have some effect in preventing peripheral conversion.



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A 32-year-old pregnant woman, with a history of a previous cesarean section (CS), is now in the third stage of labor. The delivery has been uncomplicated until now, but the third stage of labor has been prolonged, and there is difficulty in delivering the placenta. What is the most likely diagnosis?

- ☒ Abnormally adherent placenta
- ☐ Uterine atony
- ☐ Retained placental fragments
- ☐ Uterine inversion

Explanation :

Abnormally adherent placenta is the most likely diagnosis in this scenario. A history of a previous cesarean section increases the risk of placenta accreta spectrum disorders, where the placenta abnormally adheres to the uterine wall. In cases of placenta accreta, increta, or percreta, the placenta is abnormally adherent and does not detach easily during the third stage of labor, leading to prolonged third stage and difficulty in manual removal.

94

A physician colleague of yours arrives to work and smells of alcohol. Her speech is slurred and she is making repeated errors in ordering medications and diagnostic studies for patients. You are concerned about the safety of patients. Who of the following should be notified first about this physician's behavior?

- ☐ The hospital administrator
- ☐ The hospital ethics committee
- ☐ The police
- ☒ The physician

Explanation :

Health care providers have a duty to protect patients from impaired physicians when the impaired are unable or unwilling to seek help for themselves. Impairment is the inability of a provider to safely and effectively care outpatient care duties. Signs of impairment include failure to meet job responsibilities (eg, arriving late to work), forgetfulness during tasks, or evidence of substance use. The American College of Emergency Physicians (ACEP) Code of Ethics states, "Emergency physicians shall deal fairly and honestly with colleagues and take appropriate action to protect patients from health care providers who are impaired or incompetent." Handling an impaired colleague should begin in a stepwise fashion. When a colleague notices impairment of skill or judgment, he or she should approach the physician about their concerns first. If the physician fails to respond to this direct approach or patient safety or wellbeing is at risk, reporting to the other answer options may need to be utilized.

95

Which of the following tick-borne illnesses characteristically exhibits a centripetal rash?

- ☐ Lyme disease
- ☐ Babesiosis
- ☒ Rocky Mountain spotted fever



95

Which of the following tick-borne illnesses characteristically exhibits a centripetal rash?

- ☐ Lyme disease
- ☐ Babesiosis
- ☒ Rocky Mountain spotted fever
- ☐ Tularemia

Explanation :

Rocky Mountain spotted fever (RMSF), caused by *Rickettsia rickettsii* transmitted by the Dermacentor tick, causes a centripetal rash, spreading from the wrists and ankles toward the trunk. Despite its name, cases are seen most commonly in the southeastern states, but nearly all states have reported cases. The characteristic symptoms are fever, constitutional symptoms, abdominal pain, along with the centripetal rash. Antibiotic therapy (specifically with doxycycline) has steadily improved the mortality to about 1%. Doxycycline can even be given to children for optimal treatment of RMSF, as no other antibiotic improves outcomes as well. The other tick-borne diseases mentioned are rarely associated with a centripetal rash. Syphilis, dengue, Kaposi sarcoma, and coxsackievirus can also cause centripetal rashes.

96

A 3-week-old term neonate is brought by his parents with a fever of 39.0°C. He appears active and nontoxic, but his right tympanic membrane is red, and moderately swollen. The rest of the physical examination is completely unremarkable. Which of the following is the most appropriate next step in management?

- ☐ Discharge home with high-dose amoxicillin and routine follow-up.
- ☐ Discharge home with high-dose amoxicillin-clavulanic acid and next-day follow-up.
- ☐ Administer a single dose of IM ceftriaxone and discharge home with next-day follow-up.
- ☒ Admit for observation with prophylactic antibiotics and blood, urine, and CSF cultures.

Explanation :

The approach to the neonate with fever is constantly evolving. However, all neonates less than 28 days old still require a full "septic workup" with blood, urine, and CSF cultures, as well as prophylactic antibiotics and admission for observation. In this age-group, serious bacterial infection is common and often completely undetectable by physical examination or routine blood tests. Current guidelines enable the emergency physician (EP) to use considerable discretion in the evaluation of older (> 28 days) febrile neonates. It is extremely important to obtain a thorough vaccination status of young infants, since the standard vaccine schedule aggressively immunizes infants in the first year of life. Although there is no perfect algorithm for evaluating older neonates, clinicians can use a CBC to help stratify well-appearing infants > 28 days old into high- and low-risk categories for further evaluation. In addition, urinary tract infections are a common cause of occult fever among female infants as well as uncircumcised male infants. Catheterized urinalysis and culture should be sought in all febrile female infants < 24 months as well as in all uncircumcised, febrile male infants < 12 months in whom there is no alternative source of infection. The potential presence of otitis media should never result in outpatient management of the febrile neonate. Antibiotic administration without appropriate culture and laboratory analysis results in the inability to diagnose serious bacterial infection.

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Of the following, which is the strongest risk factor for an ectopic pregnancy?

- ☒ Prior ectopic pregnancy
- ☐ History of pelvic inflammatory disease
- ☐ Current use of an IUD
- ☐ Prior C-section

Explanation :

History of prior ectopic pregnancy is the strongest risk factor for current ectopic pregnancy. The most common risk factor is PID. The sequelae of PID account for approximately half of all cases of ectopic pregnancy. Although IUD use increases the relative risk of ectopic versus intrauterine pregnancy, it decreases the overall risk of all pregnancy significantly. C-section increases the risk of placenta previa, but not of ectopic pregnancy. Oral contraceptive use may reduce the symptoms of PID (thus predisposing to more untreated PID), but overall pregnancy rate (including ectopic) is reduced.

98

A 32-year-old G3P2 at 39 weeks' gestation presented to your community ED after spontaneous rupture of membranes and with regular uterine contractions roughly 3 minutes apart. Although the fetal head delivers without difficulty, the shoulders appear to be trapped and you suspect shoulder dystocia. The next best step in management should be:

- ☐ Midline episiotomy
- ☒ McRoberts maneuver
- ☐ Increased traction
- ☐ Rubin maneuver

Explanation :

Shoulder dystocia occurs when further progression of fetal delivery is halted by impaction of the fetal shoulders within the maternal pelvis. Although it is more common in diabetic mothers with infants weighing $> 4,000$ g, more than half the cases involve infants $< 4,000$ g and without other risk factors. Rapid recognition and treatment is critical because of an increased risk of fetal hypoxia and irreversible neurologic damage. Other injuries that complicate shoulder dystocia include brachial plexus injuries and humerus and clavicular fractures. With the exception of increased traction, all of the maneuvers listed may be used to help relieve shoulder dystocia. Most experts advocate using the least invasive maneuvers first. The best first step is the McRoberts maneuver in concert with suprapubic pressure. The McRoberts maneuver involves hyperflexion of the maternal hips (placing the maternal knees up to the chest), which moves the symphysis pubis over the fetal anterior shoulder. This is done in conjunction with moderate suprapubic pressure to push the fetal anterior shoulder below the symphysis. If this fails, more invasive maneuvers can be used such as the Rubin or Wood corkscrew maneuvers. The Rubin involves pushing the posterior fetal shoulder toward the fetal chest by placing a hand inside the vagina. The Wood corkscrew maneuver involves rotating the fetus 180 degrees (preferably clockwise) in an attempt to free the shoulders. In general, a liberal median episiotomy creates more space to allow the posterior shoulder to pass but does not, by itself, relieve dystocia, and it increases maternal morbidity. If all of these efforts fail, the posterior arm can be grasped, placed on the fetal chest, and swept over the face and out of the vagina. This maneuver (Barnum maneuver) may result in fractures or brachial plexus injury.



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99

Which of the following is the treatment of choice for acute cluster headache?

☒ 100% oxygen☐ Sumatriptan☐ Morphine☐ Lorazepam**Explanation :**

Oxygen is the standard of care for cluster headache. Seventy-five percent of patients with cluster headache given 100% oxygen through face mask will experience complete or near-complete relief within 15 minutes. Because attacks of cluster headaches are self-limited and typically last no longer than 90 minutes, patients may not have pain by the time they reach the ED. Therefore, oxygen is an inconvenient therapy. Sumatriptan is the most effective self-administered medication. It is administered as a nasal spray and is effective in more than 50% of patients within 15 minutes of use. However, its use is not recommended in patients who are having more than two attacks per day because this would result in an overdose of the medication. Dihydroergotamine and zolmitriptan are also effective treatments. Narcotics, benzodiazepines, and corticosteroids have no role in acute cluster headache management.

100

A 78-year-old female presents with acute onset of right-sided arm and leg weakness 1 hour prior to arrival. You suspect stroke and immediately order a noncontrast CT brain. The CT of brain is unremarkable. Which of the following is the most appropriate conclusion to draw from this CT result?

☐ There is no stroke.☐ There is no ischemic stroke.☐ There is no meningitis.☒ There is no intracranial hemorrhage.**Explanation :**

The role of noncontrast CT brain in the evaluation of an acute-onset, focal neurologic deficit is to establish the lack of hemorrhage or large space-occupying lesion in the brain. Noncontrast CT brain has poor sensitivity for ischemic stroke; because ischemic strokes represent over threequarters of all strokes, the negative predictive value of noncontrast CT brain for stroke in general is also poor. Noncontrast CT brain is used to rule out intracranial hemorrhage to permit consideration for IV tissue plasminogen activator (TPA), which is the only effective treatment for acute ischemic stroke but is absolutely contraindicated in hemorrhagic stroke. Meningitis can only be definitively diagnosed using CSF analysis with lumbar puncture. Cervical artery dissection is best evaluated with either CT angiogram or MR angiogram of the head and neck.

